

Zach Hafen-Saavedra

z.hafen.saavedra@gmail.com || zhafen.github.io || Chicago, IL ||  

Data Scientist and Manager with over 12 years of experience leading efficient solution development for complex problems, including 12 years of Python experience, 4 years of technical team leadership, 10 years of enterprise-scale relational data experience, and 6 years of natural language processing experience. Led the migration of two teams to cloud-based solutions.

Education

Northwestern University

PhD, MS, Physics and Astronomy

Specialization: Computational and Theoretical Astrophysics

2020

Evanston, IL

University of Northern Colorado

BS, Mathematical Physics

2014

Greeley, CO

Skills

Technical Skills: full-stack data science (incl. ingestion, system design, software engineering, ML/AI solution development, data engineering), healthcare analytics, NLP (incl. LLMs, vector databases, NER), relational databases, mathematical modeling and statistics

Interpersonal skills: technical leadership and management, stakeholder relations, storytelling, mentoring

Tools: Python (incl. PySpark, pytorch), Databricks (incl. asset bundles), Azure, AWS, SQL, Power BI, Streamlit, Docker, parallel computing, Git (2000+ commits/year), C/C++, and whatever gets the job done

Experience

Manager of Specialty Analytics - Research Analytics

Northwestern Medicine

June 2024 - Present

Chicago, IL

- Worked closely with the Enterprise Data, Cloud Ops, and IAM teams and Databricks support to bring our [Azure framework from conception to production-ready within a year of my hire](#).
- Subsequently trained and facilitated the migration of 20+ developers to our cloud environment, allowing us to [fully phase out usage of four legacy systems over the course of one year](#).
- Managed [five direct reports to deliver 10+ data products per week](#) to Neurology, Cardiology, Urology, and Clinical Trials institutes at Northwestern Medicine (NM).
- Organized and facilitated [weekly seminars, daily office hours, biweekly hack days, a three-day Python camp, system documentation, and a tutorial library](#), upskilling our team for a cloud environment.
- Led a 20-person roundtable and gave a 50-person talk at the Healthcare Data and Analytics Association 2025 conference, [driving dialogue about cloud infrastructure best practices, challenges, and tools](#).
- Re-envisioned from the ground up our on-premises reporting framework, designing a framework that uses Azure Databricks, Azure DevOps, Power BI, and Service Now to [deliver and maintain 1300+ active reports developed by 15+ analytics developers](#).
- Dissected the ML Engineering team's deployment framework and adapted it to our needs, creating a robust framework that is [NM's first infrastructure-as-code framework for an analytics team](#).
- Developed a [PySpark library tailored to our team's routine tasks](#), enabling analysts with deep SQL experience but limited Python experience to build reports using as little or as much Python as preferred.
- Navigated organizational and technical challenges across five teams to establish a data channel between Azure and on-prem data sources, [enabling seamless blending of cloud and legacy data](#).
- Worked closely with Identity Access Management and Cloud Security on NM's first use of B2B accounts, preparing the [infrastructure for 60+ citizen developers](#) to use our Azure resources.
- Automated [the analysis of 3000+ MySQL queries and derived database usage statistics](#) to estimate the lift to migrate to Azure, informing contractor hiring needs.
- Hired and trained analytics staff to [extend the number of supported institutes from three to four](#).
- Directed the adoption of Agile project management over the course of six months, [increasing the visibility of team efforts to over 80% of work hours](#).

- Addressed human resource issues spanning work authorization, stakeholder relations, and resource provisioning, with [all direct reports explicitly vouching that they feel supported](#).
- Fielded [10+ inquiries per week from policy to data engineering to crisis management](#), ensuring smooth operation of our team while protecting patient privacy.
- Organized [the NM NLP working group and held quarterly meetings of 15+ NLP experts](#) from across the organization, addressing gaps in NLP knowledge and establishing a community of practice.
- Collaborated with the ML/AI and Cloud Ops teams to [establish secure resources for LLM processing of sensitive data](#), enabling the use of LLMs in healthcare data analysis.
- Mentored team members in the development of an LLM-based classification model, automating the assessment of 2000+ faculty reviews and [saving thousands of hours of manual work](#).
- Instructed data engineers in re-tooling the team's medical text de-identification approach, enabling the [efficient parallel processing of 1 million+ healthcare notes](#) with improved accuracy.
- Guided analysts in the development of new named entity recognition (NER) models that draw on labeling software and LLMs to [identify 100s of patients missed by human classification](#).

Far Horizons Data Scientist

Adler Planetarium

September 2023 - June 2024

Chicago, IL

- Utilized tools including CodeBuild, Docker, ECR, and PostgreSQL to [deploy an end-to-end data analytics pipeline on AWS](#), enabling Adler staff to ingest and process data from AWS S3 with the push of a button.
- Developed an [automated ELT and image-registration pipeline](#) using Python and shell scripting, dramatically increasing the georeferencing speed from 4 manual images/hour to 5000 images/hour.
- Developed [documentation, an intuitive user interface, a suite of 40+ code tests, and a stable, containerized computing environment](#), preparing for 3 years of minimal-maintenance use by stakeholders.
- Directed the adoption of [Agile project management](#), attaining 70%+ community-driven development.
- As a museum resident scientist, [educated and collaborated with non-technical educators](#) to deliver life-changing deep-impact programs for 20+ high-school students and 4 interns.

Group Lead and McCue Prize Postdoctoral Fellow

University of California–Irvine, Department of Physics and Astronomy

July 2020 - June 2023

Irvine, CA

- Advised and mentored team members for the Dean of Physical Science's research group, leading to the advancement of [7 astrophysics and 1 computational linguistics PhD students](#).
- Developed a Python-frontend, C++-backend code to perform [NLP embeddings of scientific text](#), and presented at AI4Science on a clustering-related metric correlated with a 150%+ increase in citations.
- [Automated data retrieval from NASA APIs](#), extracting metadata for more than a million papers.
- Orchestrated a mock data challenge spanning nine international institutions, [quantifying where statistical models attained 90%+ accuracy](#), informing technical stakeholders' modeling decisions.
- Led and organized a workshop of twenty key community leaders, [fostering cross-specialty dialogue to discern high-value targets](#).

National Science Foundation Graduate Fellow in K-12 Education

Northwestern University, Department of Physics and Astronomy

June 2014 - July 2020

Evanston, IL

- Used remote resources to [apply event-tracking to 20+ TB of relational data](#), and isolated predictive parameters stakeholders could use to predict future behavior with 99%+ certainty.
- Performed [time-series decision-tree classification](#) to predict the cosmic origins of the atoms we are made of, delivering testable hypotheses to guide collaborators.
- Responsibly [used remote resources to operate 100,000-CPU-hour simulations](#), generating data used by 30+ stakeholders to increase statistical power and realism.
- Crafted [award-winning visualizations displayed throughout Chicago libraries and museums](#), advertising the beauty of science to a wide audience.
- Partnered with local schools to [pioneer a high-school data-science program](#), reaching over 100 students.
- Collaborated with 100+ researchers, leading to [36 published papers, 7 as lead author](#).